

parallel, while enabling local companies to take on more complex designs.

What parts of the chip pipeline are India's biggest bottlenecks?

Jaya: We do see progress across the pipeline, with investments and policy attention spanning the full chip pipeline. These areas are highly capital-intensive and take time to mature, which is why startups need strong ecosystem support.

Programs like the Design Linked Incentive (DLI) have helped spark interest with early IP creation, and participation from startups. But modern semiconductor development requires sustained capital, across multiple tape-outs, as design complexity continues to rise. What's needed is continued policy momentum, shared infrastructure, deeper industry participation, and funding models that reflect the true cost and timelines of building competitive chips.

What's missing on the ground that policy still doesn't solve?

Jaya: Semicon India and the Design Linked Incentive (DLI) scheme are important and timely interventions, and they have helped uplift India's design ecosystem.

Where the gap remains is in supporting the full complexity of modern chip development. The major challenges for any startup lie in these areas: access to EDA tools, IP licensing, advanced packaging, post-silicon validation, and fabrication. Each of these requires significant capital and long-term commitment.

Many startups and MSMEs struggle once they move beyond design completion.

The next phase of DLI should focus on supporting scale and complexity, helping startups move from promising designs to competitive products. Stronger industry linkages through local customers, closer integration with manufacturing partners,

and structured mentorship can help align design with real demand.

What role can India realistically play in advanced packaging and testing over the next 3–5 years?

Jaya: Packaging is emerging as the most practical opportunity for India as global demand for AI, datacentres, and automotive electronics drives the need for higher performance and energy-efficient chips. Over the next 3–5 years, India can play a focused and realistic role in advanced packaging and testing. Under the India Semiconductor Mission, approved OSAT and advanced packaging projects will give India a base to scale capabilities in areas such as chiplet integration, system-in-package and advanced testing.

To leverage this moment, India needs to do three things better: rapidly scale reliable manufacturing and process control, deepen industry partnerships for tools and materials, and tightly link packaging with its strong design ecosystem. If executed well, advanced packaging can become India's fastest route up the semiconductor value chain, positioning the country as a credible global hub for packaging and testing even without leading-edge fabs.

How do you stop India from becoming only a 'services' back-end in semiconductors and instead build IP ownership and product companies?

Jaya: To ensure India does not remain only a back-end for semiconductor services, the focus needs to shift from execution to IP ownership and product creation. India already has deep design talent; the priority now should be to convert this talent into proprietary platforms, architectures and products through sustained investment in R&D and domestic IP creation. Building and protecting IP is critical to moving up the value chain and establishing global semiconductor companies.

Alongside product development the government should also focus on market adoption. The government can play a crucial role by actively supporting and promoting made-in-India semiconductor products.

What should India do to become 'trusted silicon' provider for both domestic and global markets?

Jaya: To become a source of trusted silicon, India must build credibility across security, supply-chain assurance and global compliance, not just capacity. To compete effectively in the global market, India should integrate security-by-design throughout the entire chip lifecycle.

This process should begin with the secure development of intellectual property, and include rigorous verification and validation, ensuring that security is embedded into silicon from the very start.

As global dependencies increase, supply chain assurance becomes essential. India must enhance end-to-end traceability in design, manufacturing, packaging, and testing, supported by secure logistics, reliable partners, and appropriate protections to align with global standards and certifications.

Accredited laboratories and internationally recognized quality, safety, and security certifications are essential for global acceptance. By combining secure manufacturing, transparent supply chains and global certifications, India can position itself as a reliable and trusted silicon partner in the global technology ecosystem.

If you could ask the Indian government for three non-financial interventions, what would they be?

Jaya: India is well-positioned to develop a strong and sustainable semiconductor industry in the long term. India requires a more comprehensive integration of practical semiconductor training into academic curricula. In addition to theoretical